

1. PRODUCT OVERVIEW & CORE ARCHITECTURE

Mini is an offline personal and business finance manager engineered for small entrepreneurs and informal micro-businesses. It runs completely standalone without remote server dependencies or recurring cloud costs.

- **State Management:** Clean layer separation with provider (Repository → Provider → Screen). UI widgets contain zero business or data logic.
- **Local Engine:** sqflite with automated multi-version migrations (v1 to v4) handled in `local_database.dart`. Fully cross-compatible (Android & Windows Desktop via `sqflite_common_ffi`).
- **UI/UX:** Material Design 3 implementation with dynamic Seed Color theming, full Light & Dark mode support, and responsive data visualization dashboards.
- **Security & Authentication:** Standalone local user authentication securing master actions with salted bcrypt password hashing.
- **Localization (l10n):** Dual-language architecture (English & Portuguese) powered by standard Flutter .arb translation files switchable at runtime.

2. FUNCTIONAL CAPABILITIES

- **Multi-Store / Business Units:** Strict data segregation per store (Customers, Suppliers, Sales, and Expenses operate independently).
- **Consolidated Super Dashboard:** Aggregated multi-store metrics and financial breakdowns alongside dedicated single-store stats.
- **Flexible Sales Engine:** Instant cash sales & credit sales with automatic installment generation and sequential invoice referencing.
- **Snapshot Financial Statements:** Point-in-time exportable statement snapshots per individual branch or consolidated.

3. QUICK SETUP & ENVIRONMENT REQUIREMENTS

- **Flutter SDK:** ^3.12.1 or newer (Stable channel)
- **Target Android SDK:** compileSdk 36 (Tested and optimized on Android 12+)

To run the project in development mode:

```
# 1. Navigate to project root & fetch packages
flutter pub get

# 2. Run debug build on target device
flutter run
```

4. BACKUP & DATA RESTORATION PIPELINE

The app executes full JSON database dumps to local storage and coordinates external transfers through `share_plus`. The restoration engine performs safe line-by-line entity merging based on `updated_at/created_at` timestamps—ensuring local records are never mistakenly wiped.

5. BRANDING & CUSTOMIZATION GUIDE

A. Changing Application ID (Package Name)

Open `android/app/build.gradle.kts` and update the identifiers from the default placeholder to your unique namespace:

```
android {
    namespace = "com.yourcompany.mini"
    defaultConfig {
        applicationId = "com.yourcompany.mini"
        minSdk = 21
        targetSdk = 36
    }
}
```

B. Replacing App Icons

The package uses `flutter_launcher_icons`. Replace `assets/icon.png` with your custom 512x512 PNG asset and run:

```
dart run flutter_launcher_icons
```

6. ANDROID STORAGE PERMISSIONS & STORE COMPLIANCE

Important Notice for Google Play Publishing:

The app declares `MANAGE_EXTERNAL_STORAGE` to manage backup files directly inside the public `Downloads/Mini Backups` directory.

- **Publishing to Google Play:** You must submit the declaration form justifying broad storage access for core backup functionality, or adapt file output to app-specific directories using `path_provider` and rely exclusively on system sharing via `share_plus`.
- **Direct APK / Alternative Stores:** Standard permission granting works out-of-the-box on Android 11+ via system prompts.

7. PRODUCTION SIGNING & KEYSTORE CONFIGURATION

Before publishing or generating customer distribution builds, generate a production keystore:

```
keytool -genkey -v -keystore release-keystore.jks -keyalg RSA -keysize 2048 -validity 10000 -alias mini_key
```

Reference your keystore in `android/key.properties` and adjust `android/app/build.gradle.kts` to sign with release credentials instead of debug defaults.

8. GENERATING PRODUCTION RELEASES

Compile production binaries with tree-shaking and minification enabled:

```
# Android App Bundle (For Google Play Store)
flutter build appbundle --release --obfuscate --split-debug-info=build/app/outputs/symbols

# Standalone APK (For Direct Distribution / Demo testing)
flutter build apk --release
```